INCH-POUND

A-A-55634C
30 Oct 2019
SUPERSEDES
A-A-55634B
9 Aug 2018

COMMERCIAL ITEM DESCRIPTION

ZIPPERS (FASTENERS, SLIDE INTERLOCKING)

The General Services Administration has authorized the use of this commercial item description by all Federal Agencies.

1. Scope. This document covers slide fasteners used for dress, coveralls, aviation, rainwear, sleeping bag, equipage, body bag, chemical protective, anti-gravity (anti-g) suit, diving suit, tentage and other miscellaneous and specialty end item applications.

2. Classification: This section contains zipper configuration types/styles, descriptions and definitions.

2.1 Types/Styles: The following tables reference availability of all potential zipper configurations relative to types and styles. The referenced footnotes 1/ thru 3/ in the following tables will be noted after Table V.

TABLE I. Type I, Fastener, Nonseparating, Nonreversible

Style	Lock 1/	Top Stop	Bottom Stop	Slider Arrangement
1	Pin or Cam	Without	Without	Single
2	Automatic			
3	Without	Open	Closed	
4	Automatic			
5	Pin or Cam			
6	Without	Bridge	Closed	
7	Automatic			
8	Pin or Cam			
9	Without	Open	Open	Multiple (mouth to mouth)
10	Automatic			
11	Pin or Cam			
12	Without	Open	Bridge	
13	Automatic			
14	Pin or Cam			
15	Without	Bridge	Bridge	
16	Automatic			
17	Pin or Cam			

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TABLE I. Type I, Fastener, Nonseparating, Nonreversible - Continued

Style	Lock 1/	Top Stop	Bottom Stop	Slider Arrangement
18	Without	Closed	Closed	Multiple (throat to throat)
19	Automatic			
20	Pin or Cam			
21	Automatic	Open	Bridge	Three Sliders (mouth to mouth, throat to throat)

TABLE II. Type II, Fastener, Nonseparating, Reversible

Style	Lock 1/	Top Stop	Bottom Stop	Slider Arrangement
1	Without	Open	Closed	Single
2	Automatic			
3	Without	Bridge	Closed	
4	Automatic			
5	Without	Open	Open	Multiple (mouth to mouth)
6	Without	Bridge	Bridge	
7	Without	Closed	Closed	Multiple (throat to throat)

TABLE III. Type III, Fastener, Separating, Nonreversible

Style	Lock 1/	Top Stop	Bottom Stop	Slider Arrangement
1	Without	Open	Right-hand separating unit	Single
2				
3	Without	Open	Left-hand separating unit	
4				
5	Automatic	Right-hand quick release	Right-hand separating unit	
6	Automatic	Left-hand quick release	Left-hand separating unit	
7 7A	Automatic For Anti-Gravity Clothing use		Right-hand separating unit	
8	Automatic			

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8A	For Anti-Gravity Clothing use	Left-hand separating unit	

TABLE III. Type III, Fastener, Separating, Nonreversible - Continued

Style	Lock 1/	Top Stop	Bottom Stop	Slider Arrangement
9	Without	Right-hand quick release	Right-hand separating unit	Single
10	Without	Left-hand quick release	Left-hand separating unit	
11	Automatic	Right-hand quick release	Right-hand separating unit	Multiple (mouth to mouth)
12	Automatic	Left-hand quick release	Left-hand separating unit	
13	Automatic	Open	Left-hand separating unit	
14	Automatic	Open	Right-hand separating unit	
15	Automatic	Right-hand separating unit	Right-hand separating unit	Multiple (throat to throat)
16	Without	Left-hand separating unit	Left-hand separating unit	

TABLE IV. Type IV, Fastener, Separating, Reversible 3/

Style	Lock 1/	Top Stop	Bottom Stop	Slider Arrangement
1	Without	Open	Right-hand separating unit	Single
2	Without	Open	Left-hand separating unit	
3	Without	Right-hand quick release	Right-hand separating unit	

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4	Without	Left-hand quick release	Left-hand separating unit	

TABLE IV. Type IV, Fastener, Separating, Reversible 2/ - Continued

Style	Lock 1/	Top Stop	Bottom Stop	Slider Arrangement
5	Automatic	Open	Right-hand separating unit	Multiple (mouth to mouth)
6	Automatic	Open	Left-hand separating unit	

TABLE V. Type V, Fastener, Separating, Scoop Count 3/

Style	Lock 1/	Top Stop	Bottom Stop	Slider Arrangement
1	Without	Open	Right-hand separating unit	Single
2	Without	Open	Left-hand separating unit	
3	Without	Right-hand quick release	Right-hand separating unit	
4	Without	Left-hand quick release	Left-hand separating unit	
5	Automatic	Open	Right-hand separating unit	Multiple (mouth to mouth)
6	Automatic	Open	Left-hand separating unit	

1/ Automatic locks include ratchet type locks. Cam and pin lock include semi-automatic pin locks.
2/ Reversible sliders are those which have a pull on both the front and back of the slider or are those the pull can move from the front to the back of the slider.
3/ Scoop count type zippers are precisely manufactured individual element chain zippers based on scoop count per inch and guaranteed to engage any other scoop type fastener of same size despite different lot number or production date.

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2.2 Zipper definitions.

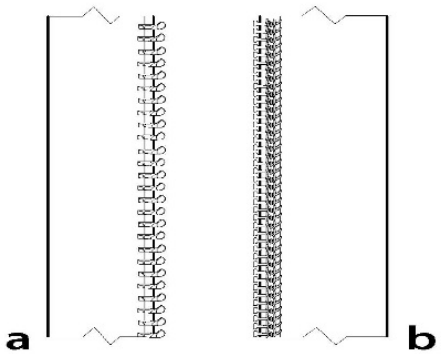
2.2.1 Chain. Chain is a continuous closure, formed by interlocking the elements or scoops on one tape alternately with those on the other tape. Existing types of chains are continuous element, individual element zippers, self-sealing and scoop count.

2.2.2 Individual element fasteners (IEF) (See Figure 1a). Two (2) rows of interlockable separately - formed teeth called scoops or elements, each attached to one (1) of the opposing edges of two (2) tapes, most commonly manufactured from, but not restricted to: brass, aluminum, acetyl plastic polymer or nylon plastic polymer. The latter polymer no.7 and 9 plastic element has two classes of chain, class 1- normal duty and class 2 - heavy duty. The heavy duty 7 and 9 IEF plastic fastener may be manufactured with nylon plastic polymer with a fin along the shoulder of the element for added strength and durability.

2.2.2.1 Water resistant IEF zippers. Plastic IEF teeth designed to resist water penetration. Fastener tape coated with a polyurethane film or similar water repellent Finish or material. A water resistant zipper possesses no resistance to hydrostatic pressure (the pressure of water at depth).

2.2.3 Continuous element fasteners (CEF) (See Figure 1b). The elements may be formed as a strand of configured monofilament or elements with integrally formed interconnecting webs or elements individually molded or fused to an interconnecting cord or cords. These rows of elements are attached either by sewing or by being integrally woven into the tape fabric. The woven model generally supplies a higher degree of abrasion resistance; however, it is best to discuss application with zipper manufacturer. CEF's are commonly manufactured from either polyester or nylon polymer.

Individual element chain type with individual sized scoop designed teeth either being metal or extruded plastic.



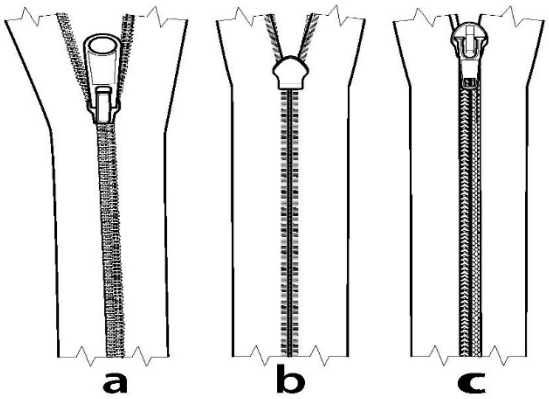
Continuous element chain type (can be sewn or woven into tape).

FIGURE 1. IEF & CEF chain elements

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2.2.3.1 Reverse continuous element fasteners (RCEF) (See Figure 2.c). RCEF is a variation of CEF where the element is to the inside of the application with the zipper tape side visible to the outside of the application. The RCEF will have a reverse slider with the pull on the zipper tape or reverse side as opposed to standard coil configuration. The RCEF may be coated with a water repellent polymer or film.

2.2.3.2 Water RCEF resistant zippers (See Figure 2.c). Fastener coated with a polyurethane film or similar water repellent Finish or material. RCEF is produced where the zipper is installed with the element to the inside of the application with the water repellent film coated tape side visible. A water resistant zipper possesses no resistance to hydrostatic pressure (the pressure of water at a depth).



a) Face side with chain up
b) Backside with abutting zipper tape
c) RCEF zipper with urethane coating and reverse slider. Zipper is water resistant not water proof.

FIGURE 2. CEF type zippers

2.2.3.3 Invisible continuous element fasteners (ICEF) (See Figure 3). ICEF is a variation of sizes 2 and 3 CEF where the element is invisible or concealed within the zipper tape when sewn into lightweight – medium applications. The ICEF cannot be produced with a separator and requires a special sewing machine presser foot (see 7.6). Number 3, CEF type zippers are hidden away within the zipper tape when sewn into lightweight to medium applications.

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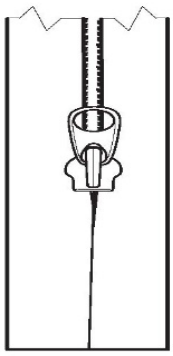


FIGURE 3. Invisible or concealed zipper

2.2.4 Zipper components (visual) (See Figures 4 thru 10).

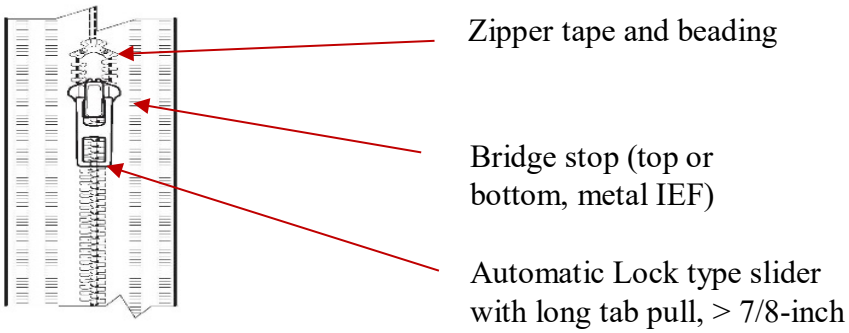


FIGURE 4. Common zipper components

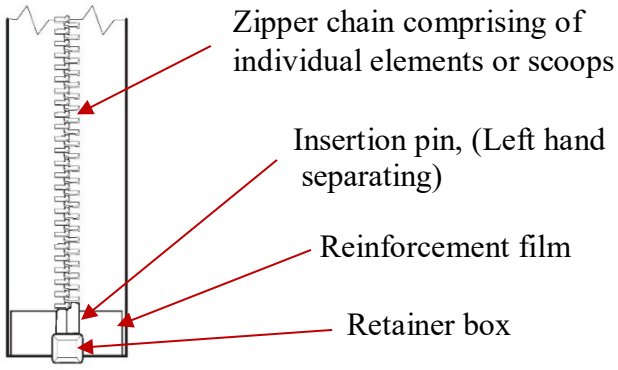


FIGURE 5. Separating zipper components (Default is Left hand Pin & Box, unless otherwise specified.)

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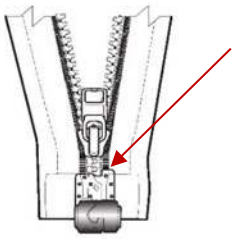


FIGURE 6. Alternate specialty separating zipper components

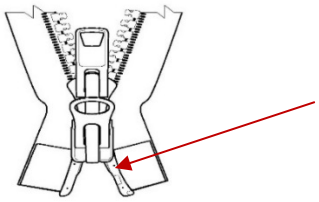


FIGURE 7. Alternate double-pin separating zipper components

2.2.4.1 Quick-release stops (See Figure 8). This device will temporarily limit the travel of the slider at the open end of the chain under normal closing operation. It will permit the slider to be forced beyond the stop when an additional force is exerted against the slider. The performance requirements for this device will be established between the design activity and the manufacturer. However, quick release stops will endure a minimum of 200 cycles when pulled (either manually or mechanically) beyond the quick release stop.

2.2.4.2 – Quick-Burst IEF – Section of zipper chain allowing for a quick opening of the fastener. IEF Type III, IV, or V zipper that has the ability to quickly separate into two (left and right) halves of the fastener. The Quick-Burst section has a lower crosswise strength compared to the standard zipper chain.

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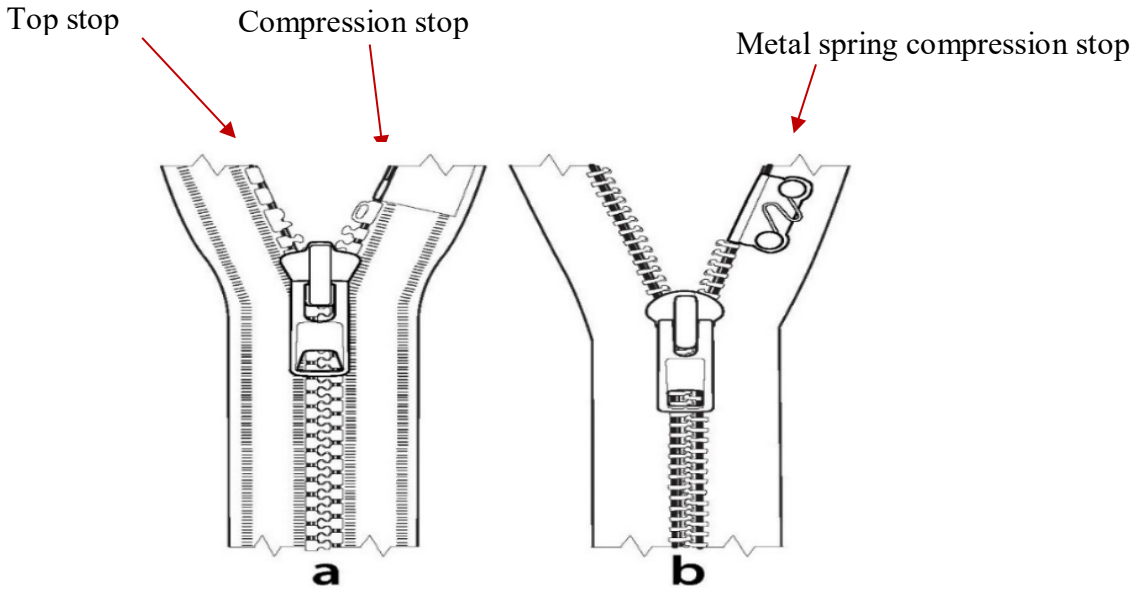


FIGURE 8. Quick Release (QR) components

a. Plastic extruded chain b. Metal Chain

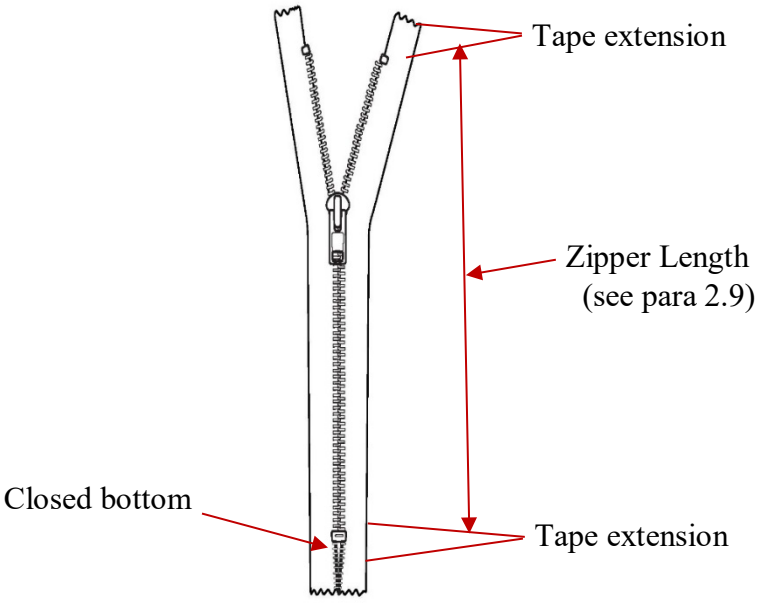


FIGURE 9. Common zipper components

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2.2.5 Self-sealing zippers. In general, self-sealing zippers consist of two (2) rows of interlockable CEF, plastic IEF, or metal IEF chain that prevents moisture penetration under hydrostatic pressure (the pressure of water at a depth), air pressure or chemical contamination from leaching in or out of zipper construction. Tapes are coated with rubber, polyurethane, polyvinylchloride, or other suitable materials. The hydrostatic pressure of the zipper shall be specified by the manufacturer or application (see 2.2.5.1 through 2.2.5.2 for various applications).

2.2.5.1 Water tight (See Figures 10 and 11). Fastener that is designed to prevent the penetration of liquid past closure's seal.

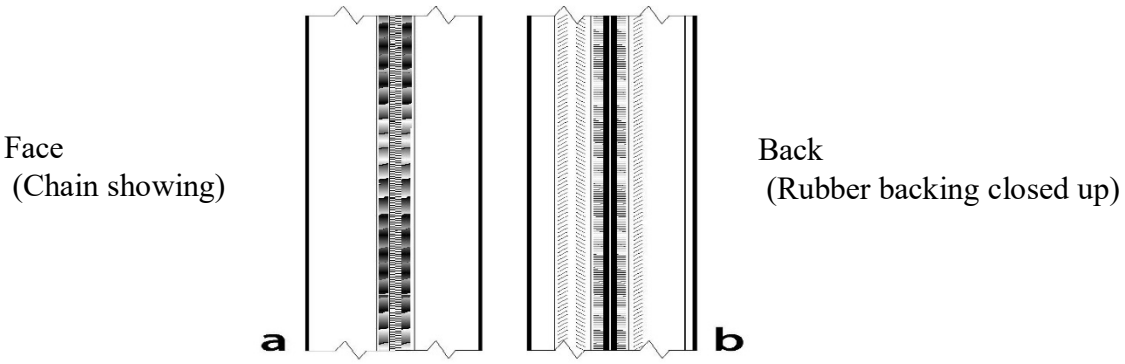


FIGURE 10. Water tight zipper

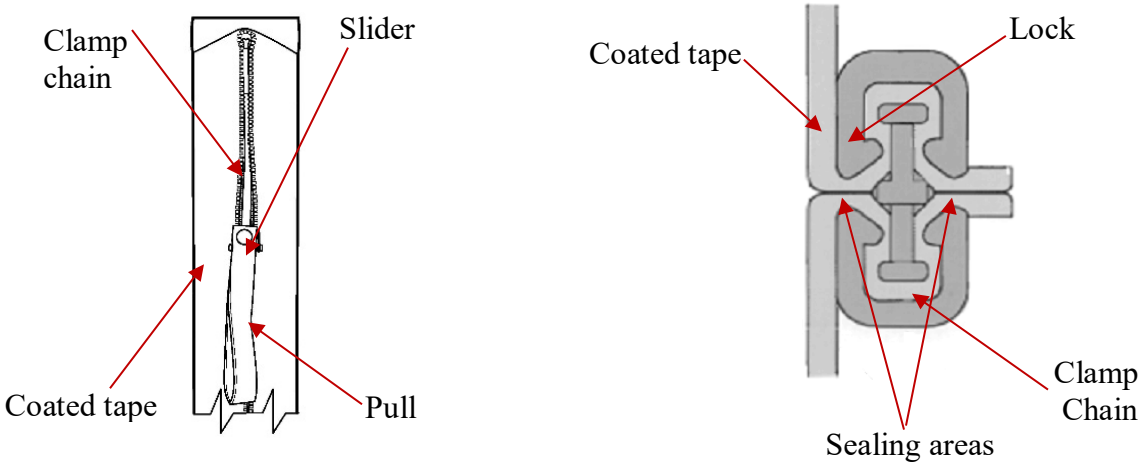


FIGURE 11. Water tight zipper components

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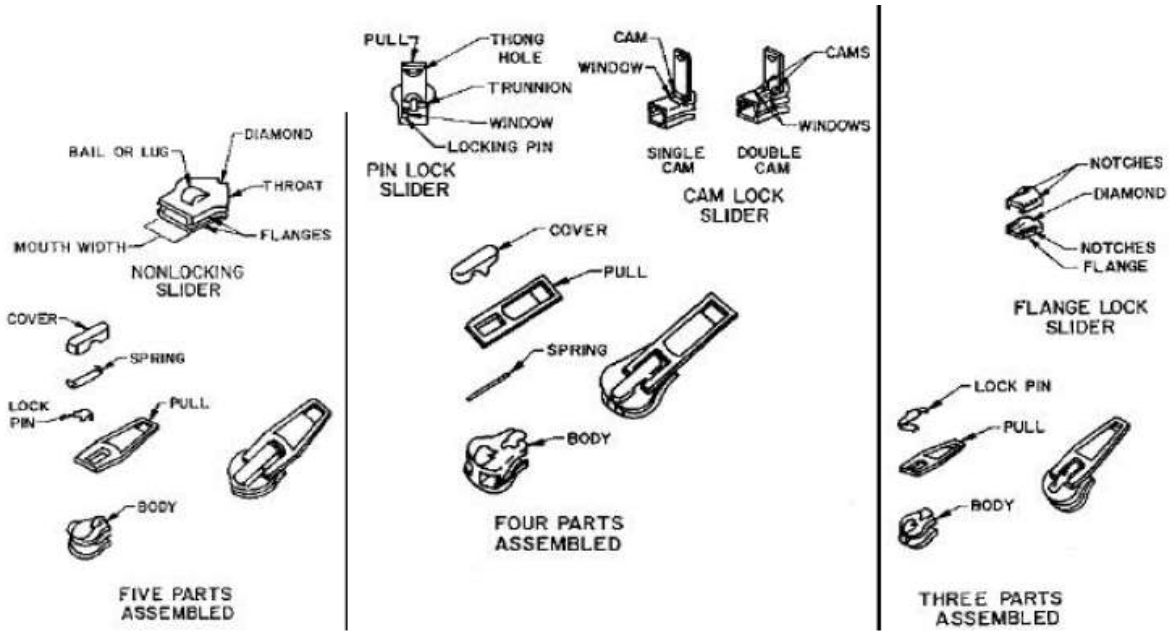
2.2.5.2 Chemical biological resistant zippers. Fasteners that are especially life critical will be resistant to specific chemical penetration requirements either in simulation swatch testing, Man in Simulated Testing (MIST), or in final live agent testing.

2.2.6 Anti-Gravity Suit zippers. Fastener generally described as size 5, 7, 8 and 10, black oxidized brass individual element fastener chain that provides a secure pin and box locking mechanism. Parts are fully functional for intended requirements and application.

2.3 Size. The zipper size is determined by combination of closed chain width and physical characteristics per Table VI. There are a total of ten (10) separate sizes within size (6) size groups, ranging from 1 through 10. Maintaining all requirements of any intra-range size products will be the responsibility of the manufacturer.

2.4 Tape. IEF zipper chains are connected/attached to tape fabrics having an overall width which yield a specific sewable area. CEF chains are connected/attached via sewn threads or woven into tape. These tapes are listed in Table VI as typical tape widths (self-sealing types are as specified). Specific tape widths are designated by the zipper manufacturer as required. Tape materials (except self-sealing) are generally polyester or aramid (defined as aramid or high temperature resistant polyamide, see 3.11).

2.5 Sliders (See Figures 12 and 13). Automatic and semi-automatic locking and non-locking sliders are movable components which span the two rows of elements comprising the chain. The slider, when moved, will open and close the zipper chain. Sliders come in either single or multiple arrangement in order to open/close in a specific manner. Sliders may be equipped with locks, these being automatic, pin, semi-automatic pin locks and cam locks. The semi-automatic pin lock slider may be interchanged for cam locks. The manufacturers name or Identification (ID) is imprinted either on the slider body or pull. Reversible sliders may have one movable pull tab or two pull tabs for use on either side of the fastener. Reverse sliders for RCEF zippers have a single pull tab for use on the reverse (tape) side of the fastener (see Figure 2.c).



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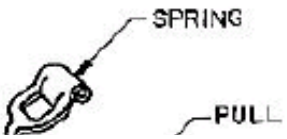


FIGURE 12. Non-locking and locking type sliders

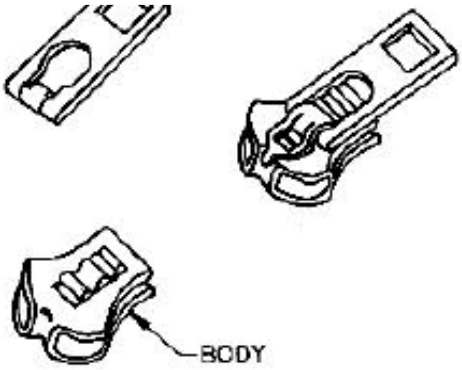
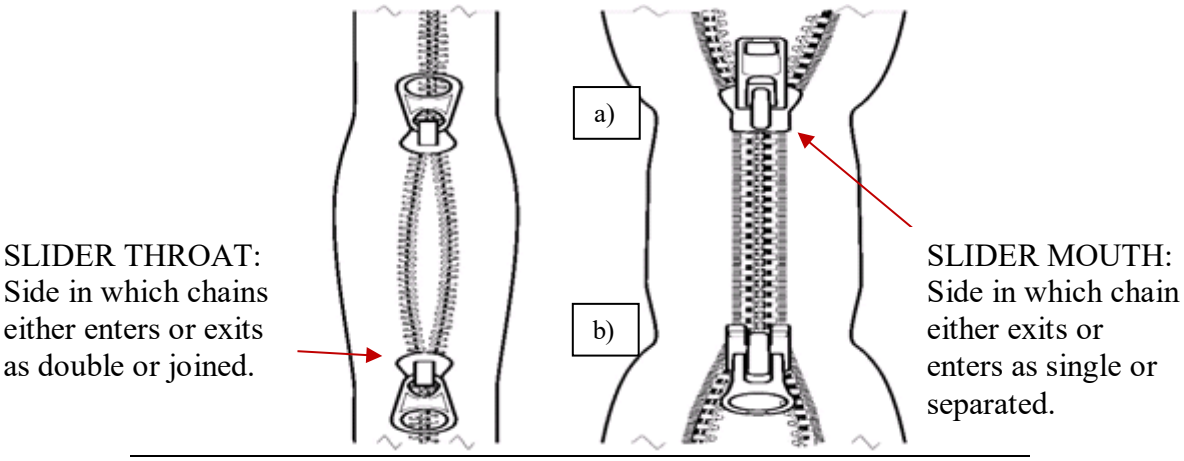


FIGURE 13. Semi-automatic locking slider

2.5.1 Slider arrangements (See Figure 14). When there is more than one slider used on a zipper then the sliders can be arranged either mouth to mouth (zipper engaged/closed between sliders) or throat to throat (zipper disengaged/open between sliders).



a) Throat to throat (zipper disengaged/open between sliders)
b) Mouth to mouth (zipper engaged/closed between sliders)

FIGURE 14. Slider arrangement

2.5.2 Slider materials. Slider bodies, pulls, and parts may be manufactured from zinc, stainless steel, plastic, brass, or other suitable materials. For salt water or electrical arc potential applications, plastic sliders are preferred. For fire resistant applications zinc, brass, or high temperature plastic sliders are suitable.

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2.6 Slider pulls (See Figure 15). Slider pulls provide the movement of the sliders. Unless otherwise specified, they can be either short (below 7/8-inch long), long (7/8-inch or longer), padlock, chain or stirrup pull. Pull arrangements are of three main types:

Non-Reversible – Opens and closes from one side.
Reversible – Opens and closes from both sides, be single pulls on both sides or a single movable pull which can slide to either side of the slider.
Reverse Single Pull RCEF – Opens and closes from the tape or back side of a RCEF zipper with a single pull on the reverse side of the slider.

A thong may be attached to a slider pull to facilitate opening of a zipper while wearing gloves. When specified, any pull that provides a hole large enough to accommodate a thong may be used or no pulls may be used with in conjunction with a thong.

2.6.1 Padlock slider body (See Figure 15). Slider which incorporates a large hole at the end of the body for the attachment of a small lock or similar security device. The ends of the body may interlock with a second slider (throat to throat).

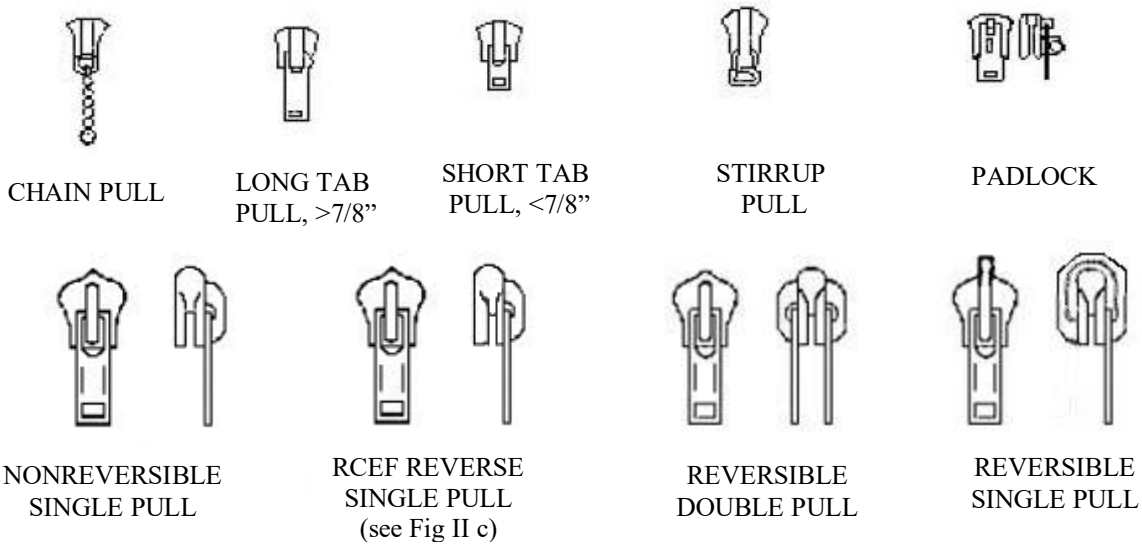


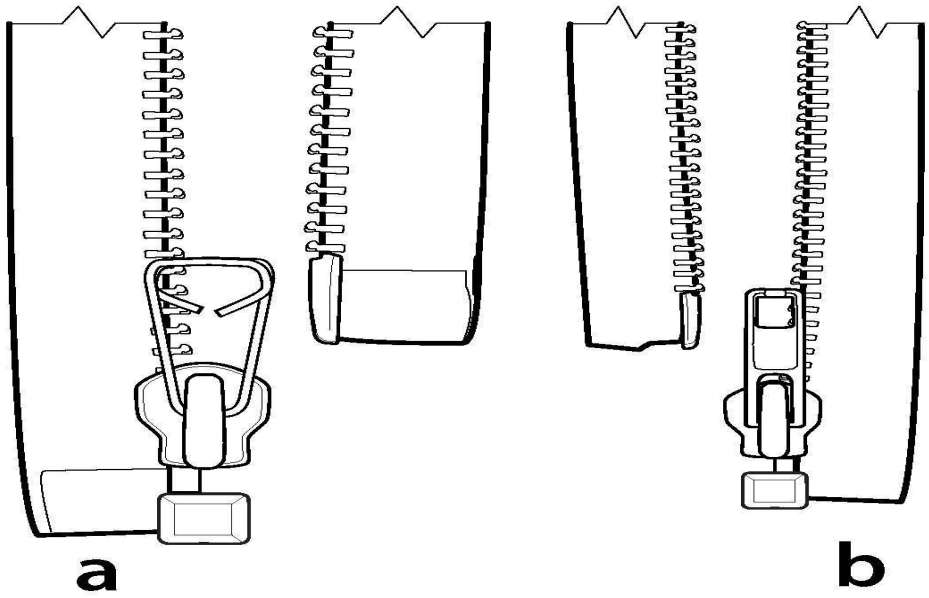
FIGURE 15. Slider pull types

2.9 Zipper length. The length of the fastener shall be measured from the bottom of the bottom stop to the top of the top stop. For self-sealing fasteners, the length is measured from the top of the bottom stop to the top of the slider when against the top stop (see Table VI).

2.10 Separating unit (hand) (See Figure 16). Separating units are used in application where two chain sides must be released from each other. All separating zippers are capable of being engaged and disengaged in a smooth even manner with minimal force. Separators come in two "Hands", left and right. To determine the "Hand" of a separator: when a separating zipper having a single pull is viewed, as worn, with the chain in a vertical position and the separating unit at the bottom of the chain, the zipper will be considered "Left Hand" if the separating pin is

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on the observers left and "Right Hand" if the separating pin is on the observers right. Unless otherwise specified, all separating units are left hand.



a) Separating unit as shown is '**Left Hand**'.
b) If the placement of the separating pin and retainer were on opposite sides, it would become '**Right Hand**'.

FIGURE 16. Separator hand

2.12 Interchangeability. An interchangeable slide fastener shall be a separating type which is constructed so that when the two stringers of the chain are separated, each stringer can be joined and operated using the slider provided by the manufacturer, with the opposing stringer of other fasteners of the same size and design from a single manufacturer to end within two (2) teeth. Neither like size interchangeable fasteners nor sliders from different manufacturers are to be considered interchangeable with each other.

3. SALIENT CHARACTERISTICS

3.1 Physical performance requirements. Table VI exhibits performance requirements relative to specific size category.

3.1.1 All components shall be from the same manufacturer; otherwise zippers may be dysfunctional

3.1.2 Tape ends/extensions. Shall be of a length that is practical for sewing, unless otherwise specified in the contract. All tape ends are treated to prevent fraying. Typical methods are: coated type, bonded reinforcement, tape impregnation, hot knife, ultrasonic sealing, or pinking. All lengths shall have a tolerance of +/- ½ inch.

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3.1.3 Thongs. Thongs are independent textile webbing/braid/outershell/Flame Resistant material extensions to the slider pulls. Finished length of the thong after assembly to the pull is 6 (+/- ¾) inches, unless otherwise stated. The thong material is 3/8-inch wide and has a breaking strength of 100 lbs. minimum. Thongs are tied to the pull. Thong ends are bartacked or treated to prevent fraying.

3.1.4 Top and bottom stops. Element or elements used to restrict the movement of the slider. Top and bottom stops may be pressed metal, molded plastic, formed metal, formed plastic or other suitable method. The use of a staple, pressed, or folded type stop cannot be used in lieu of a bridge stop, except in the case of CEFs or individual molded plastic IEF.

3.1.5 Bridge stop. Stop attached to the top or bottom of a fastener that allows the top of the slider to fully rest against the bottom of the stop. Bridge stops are only applicable to metal IEFs.

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TABLE VI. Performance Requirements

Fastener Sizes						
Physical Characteristic	1-2	3-4	5-7	8	9	10
Closed Chain Width, inches	**0.125-0.170**	**0.150-0.200**	**0.200-0.270**	**0.270-0.325**	**0.310-0.340**	**0.330-0.500**
Tape Width (± 1/16-inch/ +/-.0625)	**7/16-9/16**	**7/16-5/8**	**9/16-3/4**	**5/8-1.0**	**5/8-1.0**	**5/8-1.0**
Strength Tests: (lbs. min)						
A. Crosswise Breaking Strength of Chain:						
Individual Element Fasteners, (IEF)						
Brass/Nickel	60	70	125	190	200	225
Brass for Anti-G	-	-	190	-	250	250
Aluminum	65	65	90	100	110	125
Plastic, Class 1 Normal Duty	60	65	90	115	125	135
Class 2 Heavy Duty	-	-	135	-	250	-
Continuous Element Fasteners, (CEF/RCEF)	80	80	175	200	225	250
Water Resistant Fasteners:						
RCEF	80	80	175	200	225	250
IEF Plastic	-	-	70	-	-	100
B. Crosswise of Separating Unit:						
IEF						
Brass/Nickel/Anti-G Brass	20	20	40	40	90	100
Aluminum	20	20	30	30	75	90
Plastic, Class 1	20	20	30	30	40	50
CEF/RCEF	20	20	30	40	40	50
C. Single or Individual Element Pull-Off: (Not applicable to continuous element fasteners)						
IEF						
Brass/Nickel/Anti-G Brass	5	9	12	24	29	30
Aluminum	5	7	9	14	15	16
Plastic, Class 1	5	6	10	18	20	20
Class 2	-	-	20	-	26	-
Water Resistant Fasteners, RCEF, IEF Plastic	-	-	10	-	-	16

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TABLE VI. Performance Requirements - Continued

Physical Characteristic	Fastener Sizes 1-2	3-4	5-7	8	9	10
D. Single or Individual Element Slippage: (Not applicable to continuous element fasteners)						
IEF						
Brass/Nickel/Anti-G Brass	5	9	12	17	17	19
Aluminum	4	6	9	13	14	17
Plastic, Class 1	5	6	10	17	17	17
Class 2	-	-	15	-	22	-
Water Resistant Fasteners, IEF Plastic	-	-	10	-	-	17
E. Holding Strength of Stops, Retainers and Separating Units: (Quick Release operation over the top spring shall be 1/2 of value shown)						
IEF						
Brass/Nickel/Anti-G Brass	15	15	25	35	50	50
Aluminum	15	15	25	35	35	40
Plastic, Class 1	10	10	20	35	35	35
CEF	15	15	25	35	35	35
Water Resistant Fasteners, RCEF	15	15	25	35	35	35
Water Resistant Fasteners IEF Plastic	-	-	20	-	-	35
Operating Force Tests: (Maximum in Pounds)						
F. Slider on Chain:						
IEF						
Metal	1.5	1.5	2.5	3.5	3.5	4.5
Plastic, Class 1 & Water Repellent IEF	2.0	2.0	2.0	2.5	2.5	3.5
Class 2	-	-	2.0	-	3.0	-
CEF & Water Resistant RCEF	2.0	2.0	2.0	2.5	2.5	3.5

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TABLE VI. Performance Requirements - Continued

Physical Characteristic	Fastener Sizes 1-2	3-4	5-7	8	9	10
G. Slider on Chain after Salt Spray:						
IEF						
Metal	3.0	3.0	5.0	7.0	7.0	9.0
Plastic, Class 1 & Water Repellent IEF	4.0	4.0	4.0	5.0	5.0	7.0
Class 2	-	-	4.0	-	5.0	-
CEF & Water Resistant RCEF	4.0	4.0	4.0	5.0	5.0	7.0
H. Slider Resistance to Cushioned Compression: (Pounds to be applied)						
All fasteners	100	100	200	500	500	500
I. Operating Force after Cushioned Compression: (Maximum Operating Force)						
IEF						
Metal	1.5	1.5	2.5	2.5	3.5	4.5
Plastic, Class 1	3.0	3.0	3.0	3.0	3.5	4.0
Class 2	-	-	3.0	-	3.5	-
CEF, RCEF	3.0	3.0	3.0	3.0	3.5	4.0
J. Slider Locking Strength: (lbs. min.)						
All fasteners	3.0	3.0	7.0	7.0	7.0	10.0

3.2 Water Repellency. When designated, zipper tape shall be water repellent treated so as not to absorb more than 9.5% moisture after one laundering. Any treatment shall use only Environmental Protection Agency (EPA) approved chemicals with no toxicity (see Table VIII).

3.2.1 Water Resistant. When designated, zipper tape shall be coated with a water resistant film to prevent the penetration of water through the zipper and tape. Water penetration shall be less than 0.02 oz. when 10-inches of fastener is exposed in an angled shower test. (see 5.4).

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3.3 Colors. The color of the fasteners shall be as specified in the contract or purchase order and shall meet the requirements as specified in the document or solicitation.

3.3.1 Visual shade matching. The color of the finished tape shall match the standard sample when viewed using AATCC Evaluation Procedure 9, Option A, under filtered tungsten lamps which approximate artificial daylight D75 illuminant having a correlated color temperature of 7500 (+20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 (+ 200) Kelvin. Color of zipper components (tape, chain, sliders, thongs, stops, pin and box) shall all approximately match, except CEF, where the pin/box shall be in accordance with manufacturer's standard practice.

3.3.2 Colorfastness Requirements. Specific colorfastness requirements including chlorine bleaching, weathering, wet dry-cleaning, light, laundering and dry heat sublimation shall be designated in the procurement document. Aramid fastener tape shall only be subjected to 6 standard fading hours. All ratings (as applicable) shall be "good". See Table VIII for test methods.

3.4 Black oxidized finish. Any non-glossy (dull) black oxidized finish shall be durable with no chipping or delamination up to three (3) home launderings (see Table VIII).

3.4.1 Enameled finish. When required, metallic components shall be enameled to match the shade of the tape, unless otherwise specified in the procurement document. Anodizing or chromate finish are not allowed.

3.5 Salt spray resistance. When applicable, a 5% salt spray solution shall be exposed to the zipper for 24 hours, unless otherwise specified (see Table VIII).

3.6 Tape uniformity (straightness) of chain. Overall fastener chain shall be uniform to l/l6 of an inch and 5-inch length (see Table VIII).

3.7 Continuous chain application. Continuous chain application may be utilized as an alternate to preassembled Type I and II slide fasteners as part of the manufacturing method. This method requires that the entire chain, tape and tape extensions be securely stitched with the required seam without any puckering or waviness. The slider shall be indexed into position to provide for a smooth running chain disengagement with its mouth squarely against the bottom stop and smooth running chain engagement with its throat evenly against the top stops with no excess chain elements showing. If continuous element monofilament chain is used, sewn stops, without gapping, are allowed.

3.8 Flatness. Fastener chain shall be flat without waviness (see Table VIII).

3.9 Length and tolerances. Unless otherwise specified, the tolerances shall be as specified in Table VII. All interchangeable slide fasteners for each length of a single manufacturer's design must be made according to a uniform element account (see interchangeability) and length tolerance shall be twice those indicated in Table VII, except that the total tolerance shall be equally divided on the plus or minus side. On removable lining applications the slide fasteners

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may be center marked in accordance with user requirements. The fastener length shall be measured prior to being sewn into the suit or assembly.

TABLE VII. Length and Tolerance 1/

Fastener Size								
	1-4		5-7		8-9		10	
Length, inches	plus	minus	plus	minus	plus	minus	plus	minus
Up to 12	1/8	1/8	3/16	3/16	1/4	1/4	5/16	5/16
12-1/4 to 18	1/8	3/16	3/16	1/4	1/4	5/16	5/16	3/8
18-1/4 to 24	1/8	1/4	3/16	5/16	1/4	3/8	5/16	7/16
24-1/4 to 30	1/8	5/16	3/16	3/8	1/4	7/16	5/16	1/2
30-1/4 to 36	1/8	3/8	3/16	7/16	1/4	1/2	5/16	9/16
36-1/4 to 42	1/8	7/16	3/16	1/2	1/4	9/16	5/16	5/8
42-1/4 to 48	1/8	1/2	3/16	9/16	1/4	5/8	5/16	11/16
48-1/4 to 54	1/8	9/16	3/16	5/8	1/4	11/16	5/16	3/4
54-1/4 to 60	1/8	5/8	3/16	11/16	1/4	3/4	5/16	13/16
60-1/4 to 66	1/8	11/16	3/16	3/4	1/4	13/16	5/16	7/8
66-1/4 to 72	1/8	3/4	3/16	13/16	1/4	7/8	5/16	15/16
72-1/4 to 78	1/8	13/16	3/16	7/8	1/4	15/16	5/16	1
78-1/4 to 84	1/8	7/8	3/16	15/16	1/4	1	5/16	1-1/16
84-1/4 to 90	1/8	15/16	3/16	1	1/4	1-1/16	5/16	1-1/8
90-1/4 to 96	1/8	1	3/16	1-1/16	1/4	1-1/8	5/16	1-3/16
96-1/4 to 102	1/8	1-1/16	3/16	1-1/8	1/4	1-3/16	5/16	1-1/4
102-1/4 to 108	1/8	1-1/8	3/16	1-3/16	1/4	1-1/4	5/16	1-5/16
108-1/4 to 114	1/8	1-3/16	3/16	1-1/4	1/4	1-5/16	5/16	1-3/8
114-1/4 to 120	1/8	1-1/4	3/16	1-5/16	1/4	1-3/8	5/16	1-7/16
120-1/4 to 126	1/8	1-5/16	3/16	1-3/8	1/4	1-7/16	5/16	1-1/2
126-1/4 to 132	1/8	1-3/8	3/16	1-7/16	1/4	1-1/2	5/16	1-9/16
132-1/4 to 138	1/8	1-7/16	3/16	1-1/2	1/4	1-9/16	5/16	1-5/8
138-1/4 to 144	1/8	1-1/2	3/16	1-9/16	1/4	1-5/8	5/16	1-11/16
144-1/4 to 150	1/8	1-9/16	3/16	1-5/8	1/4	1-11/16	5/16	1-3/4
150-1/4 to over	1/8	2/	3/16	2/	1/4	2/	5/16	2/

1/ Unless otherwise specified, all dimensions are in inches.
2/ For fasteners over 150 and 1/4-inches long, the minus shall be 1 and 13/16-inches, + 1/16-inch for each additional 6-inch increments in length.

3.10 Endurance testing. Fastener chain and quick release stops shall be tested for endurance per specification requirements. Unless otherwise specified, all fastener chain shall meet a minimum

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endurance of 500 cycles back and forth, and quick release shall meet 200 cycles. For typical endurance machine description, requirements, and settings as required for chain testing (see 7.8).

3.11 Thermal resistance testing. Aramid or equivalent tape and components (high temperature polyamide or equivalent) shall continue to function after exposure to 500°F for a period of five (5) minutes. All components of the thermal resistant zipper shall be tested (see Table VIII).

3.12 Hydrostatic pressure testing. Self-sealing zipper shall be tested by the manufacturer in a manner to ensure the zipper will prevent moisture penetration under hydrostatic pressure. The level of hydrostatic pressure a self-sealing zipper can withstand will be determined by the manufacturer or application (see Table VIII).

TABLE VIII. Test Methods

Characteristic	Requirement Reference	Test method
Tape Width	TABLE VI	ASTM D 2060
Water Repellency	3.2	FED-STD 191A, Method 5502
Water Resistant	3.2.1	AATCC 35 or 42; Type II tester, sample size is 3 zippers per lot, do not use blotter paper, weigh zipper before and after testing. The difference before and after testing on average should not be more than 0.02 ounces
Crosswise Breaking Strength:		
Chain	TABLE VI	ASTM D 2061
Separating Unit	TABLE VI	ASMD D 2061
Holding Strength of Stops:		
Bottom Stop	TABLE VI	ASTM D 2061
Separable Pin	TABLE VI	ASTM D 2061
Fixed Retainer	TABLE VI	ASTM D 2061
Top Stop	TABLE VI	ASTM D 2061
Slider locking strength (180 degrees)	TABLE VI	ASTM D 2061
Operating Force:		
Slider on Chain, Opening and Closing	TABLE VI	ASTM D 2062
Slider on Chain After Salt Spray (if applicable)	TABLE VI	ASTM D 2062
Element Pull-Off		ASTM D 2061
Element Slippage –	TABLE VI	

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lengthwise		ASTM D 2061
Slider: Cushioned Compression length Salt Spray Resistance (5% for 24 hours) Uniformity (straightness) Flatness	TABLE VI 3.5 3.6 3.8	 ASTM D 2061 ASTM D 2059 ASTM D 2060 ASTM D 2060 ASTM D 2060
Endurance Testing	3.10	7.3
Thermal Resistance: Resistance to Heat	3.11	NFPA 2112 Section 8
Hydrostatic pressure testing	3.12	7.5

TABLE VIII. Test Methods - Continued

Characteristic	Requirement Reference	Test method
Colorfastness Tests	as appropriate	AATCC
Chlorine Bleaching		Test Method 3
Weathering		Test Method 111 B (20 hrs.)
Wet Dry Cleaning		Test Method 132
Light		Test Method 16 20 KJ for 20 hrs.
Laundering/ Black Oxidized Finish Durability (3x)		Test Method 61 (3A)
Dry Heat Sublimation		Test Method 117

3.13 Toxicity. The finished fastener shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 5.4. Chemicals recognized by the Environmental Protection Agency (EPA), Occupational Safety and Health Administration (OSHA) or International Agency for Research on Cancer (IARC) as human carcinogens shall not be used.

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3.14 Endurance Test requirements (See section 7.8 Endurance test).

FASTENER SIZE	FORCE, LATERAL LBS. (KGS)	FORCE, LONGITUDINAL LBS. (KBS)
1-2	1.5 (0.7)	1.1 (0.5)
3-4	2.2 (1.0)	1.5 (0.7)
5-7	3.5 (1.6)	3.1 (1.4)
8-9	5.3 (2.4)	4.0 (1.8)
10	6.6 (3.0)	5.1 (2.3)

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCTION CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards and quality assurance practices and the same product shall be offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Visual examination. Any hole, cut, or tear; color not as specified; any part shaded; any streaks; any spot or stain (topside); raw edges of tape; width of tape, zipper/fastener or selvages not as specified; any missing or broken yarn, teeth, scoops or elements shall be considered a defect.

5.3 Acceptance criteria. Acceptance criteria and lot size shall be as specified in the contract or purchase order.

5.4 Toxicity test. When required (see 7.5), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of the studies indicate the zipper/fastener is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 7.2.3). If the toxicity requirement (see 3.13) can be demonstrated with historical use data, toxicity testing may not be required.

6. PACKAGING

6.1 Packaging. Unless otherwise specified, preservation, packing and marking shall be specified in the contract or purchase order.

7. NOTES

7.1 Source for Government documents.

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7.1.1 Federal Acquisition Regulations (FARs) are available online at http://acquisition.gov/far/index.html or by contacting the Superintendent of Documents at 202-512-1800.

7.2 Sources for Non-Government documents.

7.2.1 AATCC Standards are available online at http://www.aatcc.org or from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.

7.2.2 ASTM Standards are available online at http://www.astm.org or from ASTM INTERNATIONAL, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.

7.2.3 "Hayes' Principles and Methods of Toxicology (Sixth Edition),edited by A. Wallace Hayes and Claire L. Kruger, is available online at http://www.taylorand francis.com or from Taylor & Francis Group, 6000 Broken Sound Parkway NW, Suite 300, Boca Raton, FL 33487-2742." This 6th edition was released in 2014. Another source is http://www.crcpress.com.

7.2.4 NFPA 2112 Section 8 is available online at http://www.nfpa.org or from the National Fire Protection Association, 11 Tracy Drive, Avon, MA 02322 or National Fire Protection Association, 1 Battermarch Park, Quincy, MA 02269.

7.3 Part numbering system. For purposes of fastener identification, use the following system:

Example: AA55634 - 00I 01 CEF A FS P 25
Whereby: AA55634 = Basic CID No.

00I = Type Fastener: (See Tables I-IV for types)
- 00I = Fastener non-separating, non-reversible
- 0II = Fastener non-separating, reversible
- III = Fastener separating, non-reversible
- IV = Separating reversible
- V = Scoop count

01 = Style, 01 thru 20 (See Tables I-IV for styles)

CEF = Chain Type: CEF = Continuous Element Fastener
- IEF = Individual Element Fastener
- SSF = Self-Sealing Fastener
- AGF = Anti-Gravity Fastener
- RCEF = Reversible Continuous Element Fastener
- HCEF = Hidden Continuous Element Fastener

A = Chain Material: A = Aluminum C = Brass, Black Oxidized

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B = Brass

E = Aluminum, Enameled finish
N = Nickel
P = Plastic
H = Plastic, Thermal Resistant

FS = Fastener Size: 01 = 1-2
03 = 3-4
05 = 5-7
08 = 8
09 = 9
10 = 10

P = Tape Material: P = Polyester, without water repellent (WR) finish
Q = Polyester, with WR finish
R = Polyester with water resistant film coating
N = Aramid (high-temperature-resistant polyamide) without WR finish
O = Other material as specified in the procurement document.

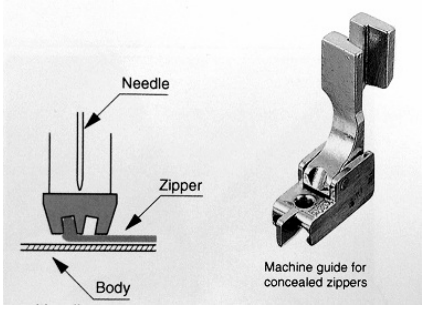
25 = Length in ¼-inches, (25 = 25/4 = 6/1/4")*.
**NOTE: Manufacturers are permitted to use either the direct length or the quarter inch system.*

7.4 Standard samples. For standard samples address the contracting activity issuing the invitation for bids or request for proposal.

7.5 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this Commercial Item Description.
b. Type and Style required (see 2.).
c. Size, width and color of fastener required (see 3.1).
d. Length and tolerances required (see 3.9).
e. When toxicity testing is required (see 3.13).
f. Product conformance provisions (see 5.1).
g. Packaging required (see 6.1).

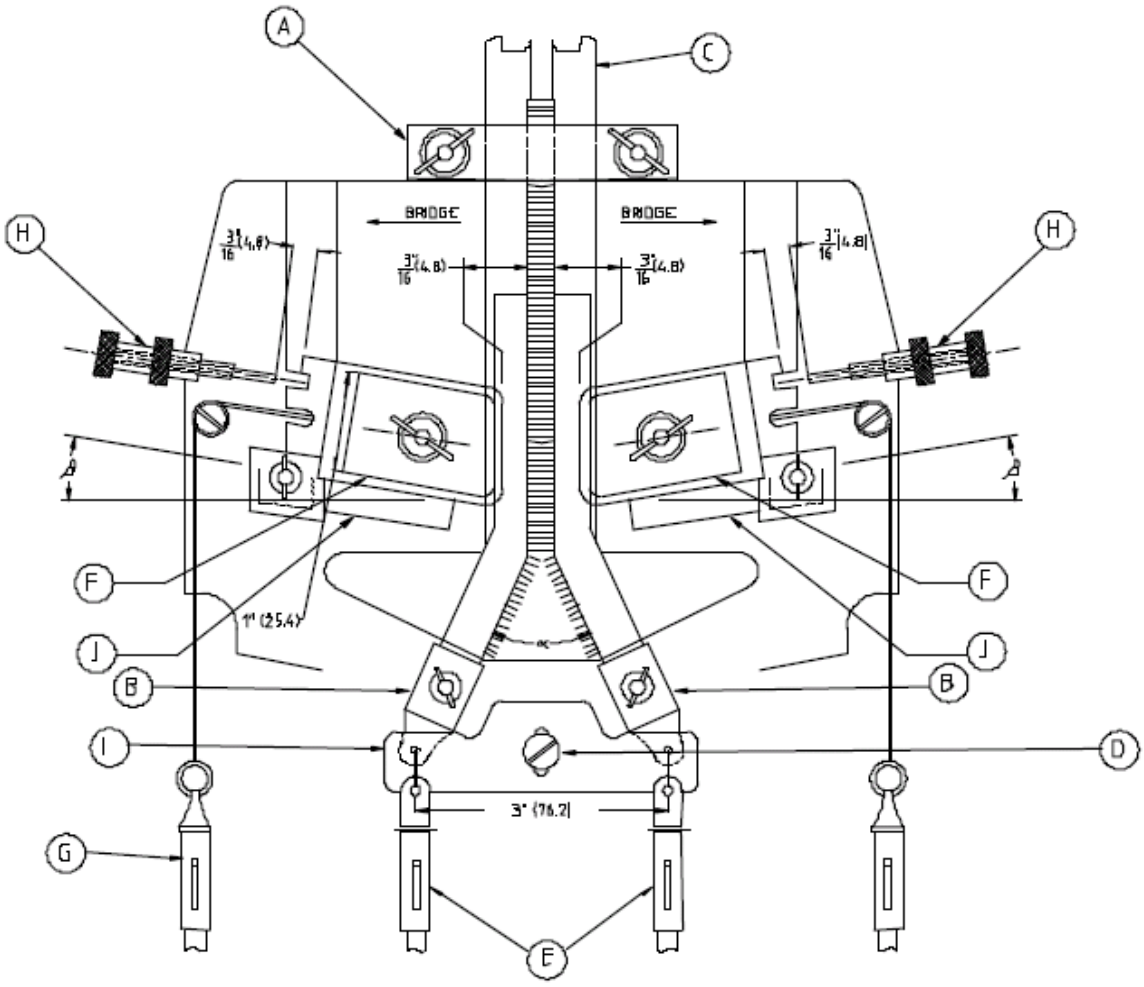
7.6 Invisible zipper presser foot. Typical sewing 301 lockstitch sewing machine presser foot design for attaching invisible zippers.



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7.7 Chain width. The closed chain width shall be measured to determine conformance of fastener size with the following procedure: Place the closed slide fastener or slide fastener chain on a horizontal working surface. Position the identification gauge with the outside faces of the anvils in flat contact with the tape. Close the jaws of the gauge slowly until they are in contact in a relaxed state, without physically compressing the shoulders of the elements and parallel to them on both stringers. Make five (5) measurements on each specimen not less than l-inch apart, reading the dial of the gauge to the nearest .001-inch.

7.8 Endurance testing. This apparatus is an endurance testing machine which attempts to simulate various applications for different slide fasteners by applying a range of lateral and longitudinal forces to the chain.



A - clamp for closed end of slide fastener
B - clamp for open end of slide fastener

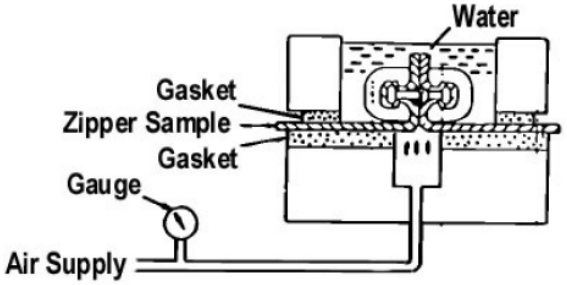
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C - sample slide fastener
D - holding pin
E - spring balance for longitudinal direction
F - clamp for holding lateral side of slide fastener
G - spring balance for lateral direction
H - adjusting screw for moving distance of lateral clamp
I - rotary plate which acts as a medium for forces between slide fastener and balance
J - supporting plate for lateral holding clamp

- Two lateral clamps are located in the midst of reciprocating stroke on both sides of slide fastener.
- Gross weight of longitudinal clamps including B and I is 300 grams.
- Sample position 3 inches distance between the pivot of B, angle open (30 degrees), angle closed (60 degrees)
- Lateral clamps are inclined to be 10 degrees at the right to the chain
- Metric dimensions are noted in parentheses.
- Endurance machine requirements are as follows:

A - the reciprocating stroke shall be 3-inches
B - unless otherwise specified, the number of cycles back and forth shall not be less than 500.
C - the lateral and longitudinal forces applied shall be in accordance with chain force table below for the different fastener sizes.
D - the fastener shall be opened and closed at a uniform rate of 30 cycles per minute.
E - the slider shall move approximately 3 inches for opening and 3-inches for closing.
F - there shall be no interference by any locking device on the slider during the opening and closing of the fastener.
G - for fasteners with multiple sliders, each shall be tested individually

7.9 Hydrostatic pressure testing.



- This apparatus is for the hydrostatic pressure testing of self-sealing zippers.

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- The zipper sample must be sealed on sides, top, and bottom to prevent air leakage.
- Bottom and top stops shall also be tested if required by application.
- Water is placed over the entire zipper, covering the zipper completely to a depth of approximately 1-inch.
- Air pressure is determined by the manufacturer, requirements or application.
- The top of the zipper is observed for air leaks (bubbles).
- Test is a pass/fail at required pressure for application.
- Testing is only conducted by the manufacturer. Certificate of compliance shall be required. There are no commercial test methods for this determination.

Keywords:

1. anti-gravity (anti-g)
2. apparel
3. aviation
4. body bag
5. chain
6. chemical protective
7. coverall
8. diving suit
9. dress
10. equipage
11. pull
12. rainwear
13. sleeping bag
14. stirrup
15. tab
16. tentage

Conversion of VF-106 parts: VF-106 vs. A-A-55634 Conversion Tables are appended at the end of the document.

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Appendix A

Comparison Table A-1 – Chain widths, V-F-106 vs. A-A-55634 (CID)

V-F-106	Sizes	Chain Width inches	CID Sizes	CID Chain width , inches	V-F-106 to CID equivalents
L	Light	0.125-0.180	1-2	0.125-0.170	L, LS = 1-2
LS	Light Special	0.125-0.180			
LM	Light Medium	0.180-0.310	3-4	0.150-0.20	LM. LMS = 3-4, 5-7, 8
LM5	Light Medium Special	0.180-0.310			
LX	Light X	0.125-0.180			LX = 1-2
M	Medium	0.180-0.310	5-7	0.200-0.270	M, MS = 3-4, 5-7, 8
MS	Medium Special	0.180-0.310			
MH	Medium Heavy	0.310-0.390	8	0.270-0.325	MH, MHS = 8. 9. 10
MHS	Medium Heavy Special	0.310-0.390	9	0.310-0.340	
H	Heavy	0.32-0.500	10	0.310-0.340	H = 9, 10

Comparison Table A-II – Break Strength equivalents

V-F-106		Brass		Aluminum		Plastic		Continuous Element (Coil)	
		V-F-106	CID	V-F-106	CID	V-F-106	CID	V-F-106	CID
L	Light	66	(1-2) 60	55	65	60	60	85	80
LS	Light Special	70	(1-2) 60	65	65	65	60	85	80
LM	Light Medium	80	(3-4) 70	75	65	75	65	175	80
LMS	Light medium Special	100	(3-4) 70	90	65	85	65	175	80
LX	Light X	130		120		90		100	
M	Medium	130		120		100		175	
MS	Medium Special	190		150		110		175	
MH	Medium Heavy	210		175		130		250	
MHS	Medium Heavy Special	250		225		145		250	
H	Heavy	275		250		160		270	

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Comparison Table III – V-F-106 conversion to CID size –

V-F-106	Brass		Aluminum		Plastic		Continuous Element (Coil)	
L	~~CID 3-4~~	CID 1-2	CID 1-2		CID 1-2		CID 1-2	
LS	~~CID 3-4~~	CID 1-2	CID 1-2		~~CID 3-4~~	CID 1-2	CID 1-2	
LM	~~CID 3-4~~		~~CID 1-2~~	CID 3-4	CID 3-4		~~CID 5-7~~	CID 3-4
LMS	CID 5-7	CID 3-4	~~CID 5-7~~	CID 3-4	~~CID 5-7~~	CID 3-4	~~CID 5-7~~	CID 3-4
LX	~~CID 5-7~~	CID 1-2	~~CID 9~~	CID 1-2	~~CID 5-7~~	CID 1-2	~~CID 5-7~~	CID 1-2
M	~~CID 5-7~~		~~CID 10~~	CID 5-7	~~CID 8~~	CID 5-7	CID 5-7	
MS	~~CID 8~~	CID 5-7	~~CID 10~~	CID 5-7	~~CID 8~~	CID 5-7	~~CID 10~~	CID 5-7
MH	~~CID 9~~	CID 8	~~CID 10~~	CID 8	~~CID 9~~	CID 8	~~CID 10~~	CID 8
MHS	~~CID 10~~	CID 9 or 10	~~CID 10~~	CID 9 or 10	~~CID 10~~	CID 9 or 10	~~CID 10~~	CID 9 or 10
H	~~CID 10~~	CID 9 or 10	~~CID 10~~	CID 9 or 10	~~CID 10~~	CID 9 or 10	~~CID 10~~	CID 9 or 10

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Comparison Table A-IV Fed Spec V-F-106F vs. CID A-A-55634 Cross-Reference of TYPES and STYLES

Type I - Nonseparating, nonreversible		Type II - Nonseparating, nonreversible, curved		Type III - Nonseparating, reversible		Type IV - Separating, nonreversible		Type V - Separating, reversible	
V-F-106 style	CID style	V-F-106 style	CID style	V-F-106 style	CID style	V-F-106 style	CID style	V-F-106 style	CID style
1	1	1	NA	1	1	1	1	1	1
1A	2	2	NA	2	2	2	NA	2	NA
2	3	3	NA	3	3	3	3	3	2
3	4	4	NA	4	4	4	NA	4	NA
4	5	5	NA	5	5	5	5	5	3
5	NA	6	NA	6	7	6	6	6	4
6	6	7	NA	7	6	7	7	7	5
7	7	8	NA			7A	7A	8	6
8	8					8	8		
9	NA					8A	8A		
10	9					9	NA		
11	10					10	NA		
12	11					11	9		
13	NA					12	10		
14	12					13	11		
15	13					14	12		
16	14					15	13		
17	NA					16	14		
18	18								
19	19								
20	20								
21	NA								
22	15								
23	16								
24	17								
25	NA								

NA- Lock types Pin, Cam and Flange deleted in CID. Replace with Automatic type lock. V-F-106, Type II, curved no longer available.

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MILITARY INTERESTS:

Custodians
Army-GL
Navy – AS
Air Force – 11

Review Activities
Army – AV, MI
Navy – SA, OS, NU

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA – FSS
VA – OSS
USDA – AFS

PREPARING ACTIVITY:

DLA – IS

Project 5325-2020-001

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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